Software Requirement Specification (SRS)

For Risk Analytics & Back-testing System

1. Introduction

1.1 Purpose

The **Risk Analytics & Back-testing System (RABS)** is designed to help portfolio managers, traders, risk managers, and quantitative researchers monitor risk, analyze portfolio exposures, test trading strategies, and optimize execution performance.

This system will:

- Deliver **daily risk metrics** such as Value at Risk (VaR) and Expected Shortfall (ES) to help risk managers monitor exposure.
- Provide a **backtesting framework** that supports momentum and mean reversion strategies with realistic transaction cost modeling.
- Generate **portfolio optimization recommendations** under multiple risk and return constraints.
- Offer **transaction cost analysis (TCA)** to help reduce slippage and market impact.
- Produce **daily risk & performance reports** that are audit-ready.

This SRS document serves as:

- A guide for developers and testers.
- A reference for stakeholders to validate that the system meets requirements.

1.2 Scope

The system aims to be a **comprehensive risk and analytics platform** for a single team or firm.

In Scope:

- Computation of portfolio and strategy-level VaR/ES at 95% and 99% confidence.
- Factor exposure breakdowns (industry, sector, style factors).
- Stress testing scenarios (rate shocks, volatility spikes, sector drawdowns).
- Breach alert generation when exposures exceed defined thresholds.
- Backtesting of trading strategies with walk-forward validation.

- Transaction cost modeling including commission, slippage, and market impact.
- Position sizing and risk constraints during back-tests.
- Portfolio optimization under turnover, sector, and risk budget limits.
- TCA reports attributing P&L to alpha, beta, cost, and timing.
- Daily performance reports with net exposures, Sharpe ratio, and drawdowns.
- Data and configuration versioning for reproducibility.

Out of Scope:

- Real-time order execution and live trading.
- Direct connection to brokers for order routing.
- Market prediction or alpha generation beyond strategy back-tests.
- Tax reporting or accounting functionality.

1.3 Definitions, Acronyms, and Abbreviations

- **VaR (Value at Risk):** Statistical measure that estimates the maximum loss over a given time horizon at a given confidence level.
- **ES (Expected Shortfall):** Also called Conditional VaR, the average loss given that the VaR threshold is breached.
- **TCA (Transaction Cost Analysis):** Study of realized trading costs such as slippage and market impact.
- **P&L:** Profit and Loss.
- **Sharpe Ratio:** Risk-adjusted performance metric = (Portfolio Return - Risk-Free Rate) / Portfolio Volatility.
- **Back-testing:** Simulation of strategy performance on historical data.
- **Walk-Forward Validation:** Testing model performance by rolling through time with re-training and re-evaluation.

1.4 References

- Industry standard risk model documentation (e.g., Barra, Axioma).
- Quantitative finance books for VaR, ES, and portfolio optimization.
- Existing TCA methodologies and benchmarks.

1.5 Overview

The rest of this document covers:

- **Section 2:** Overall description, including user characteristics, constraints, and assumptions.
- **Section 3:** Detailed functional requirements, performance criteria, and design constraints.
- **Section 4:** Possible future enhancements.
- **Appendices:** Sample formulas and report templates.

2. Overall Description

2.1 Product Perspective

RABS is a **standalone analytics platform** that runs on a secure server or cloud environment. It must:

- Integrate with market data feeds (end-of-day prices, factor returns).
- Process daily trading and holdings data from internal databases.
- Run risk and backtesting jobs automatically as scheduled batch processes.
- Provide dashboards and reports for end users.

2.2 Product Functions

The system will support the following major functions:

- **Daily Risk Analytics Job:** Compute VaR, ES, factor exposures, and generate breach alerts.
- **Backtesting Framework:** Simulate strategies, model costs, and ensure reproducibility.
- **Portfolio Optimization:** Suggest target weights under constraints.
- **Transaction Cost Analysis:** Attribute costs and recommend improvements.
- **Reporting:** Generate daily PDF/Excel reports and provide interactive dashboards.

2.3 User Characteristics

- **Risk Managers:** Require timely risk metrics and alerts.
- **Portfolio Managers:** Need optimization recommendations and performance reports.
- **Quants/Researchers:** Use backtest results and factor analyses to refine strategies.
- **Compliance/Audit Teams:** Need reproducible, audit-ready outputs.

2.4 Principal Actors

- **User:** Risk manager, PM, quant, or analyst who interacts with the system.
- **System:** The analytics platform executing jobs, storing data, and generating reports.
- **Market Data Feed:** Provides pricing, factor returns, and benchmarks.

2.5 General Constraints

- System must complete daily risk job **before market open**.
- Must handle multiple portfolios simultaneously.
- Must be secure and auditable, with access controls.

2.6 Assumptions and Dependencies

- Market data is delivered on time and is of high quality.
- Compute resources (cloud/cluster) are available for risk calculations.
- Users have necessary permissions and credentials to access the system.

3. Specific Requirements

3.1 Functional Requirements (Use Cases)

Use Case 1: Run Daily Risk Analytics

Primary Actor: System
Precondition: End-of-day market data and portfolio holdings available.
Main Scenario:

1. Scheduler triggers daily risk job.
2. System loads portfolio positions and prices.
3. System computes VaR and ES at 95% and 99% confidence levels.

4. System computes factor exposures and stress test results.
5. System stores results in risk database.
6. System generates alerts for any breaches.
7. System produces a risk report.

Alternate Scenario:

- If data is missing or corrupted, system halts job and notifies risk team.

Use Case 2: Back-test Strategy

Primary Actor: Quant/Researcher
Precondition: Historical data and strategy configuration provided.
Main Scenario:

1. User uploads strategy config (momentum or mean reversion).
2. System runs simulation with walk-forward validation.
3. System models transaction costs and applies risk constraints.
4. System generates performance metrics (return, Sharpe, max drawdown).
5. Results stored with unique ID for reproducibility.

Use Case 3: Portfolio Optimization

Primary Actor: Portfolio Manager
Precondition: Current holdings and constraints provided.
Main Scenario:

1. System receives optimization request.
2. Computes optimal weights under risk, sector, and turnover constraints.
3. Generates recommended trades to move from current to target weights.
4. Produces expected risk/return summary.

Use Case 4: Generate TCA Report

Primary Actor: System
Precondition: Execution data available.
Main Scenario:

1. System calculates realized slippage and market impact.

2. Attributes P&L into alpha, beta, cost, and timing components.
3. Produces weekly TCA report and sends to PMs and execution desk.

Use Case 5: Generate Daily Report

Primary Actor: System
Precondition: Risk, performance, and exposure data available.
Main Scenario:

1. System consolidates VaR/ES, factor exposures, drawdowns, and Sharpe ratio.
2. Includes TCA highlights and exceptions.
3. Links metrics to data snapshot and code version for audit.
4. Publishes report in PDF/Excel and dashboard format.

3.2 Performance Requirements

- Daily risk job must finish within **30 minutes** after data is available.
- Must scale to **10,000+ positions** and multiple strategies.
- 95% of user queries must return results within **2 seconds**.

3.3 Design Constraints

- **Security:** All data must be encrypted at rest and in transit.
- **Reproducibility:** Configs, seeds, and versions must be logged.
- **Fault Tolerance:** Must handle node failures without data corruption.

3.4 External Interface Requirements

- **Dashboards:** Web interface to view reports interactively.
- **APIs:** REST endpoints for quants to fetch data.
- **Reports:** PDF/Excel exports for compliance and audit.

4. Future Extensions

- Real-time intraday risk monitoring.
- Support for additional asset classes (options, futures, crypto).

- Machine learning models for cost prediction and dynamic risk limits.

5. Appendix

Appendix A: Formula for VaR

VaR = Quantile of loss distribution at given confidence level.

Appendix B: Formula for Expected Shortfall

ES = Average loss exceeding VaR threshold.

Appendix C: Sample Report Layout

- **Risk Summary Table:** VaR, ES, factor contributions.
- **Performance Section:** Returns, Sharpe, drawdowns.
- **TCA Section:** Cost attribution and recommendations.